# Response to the Action Editor and Reviewers

Manuscript: *Difficulty Estimation, Simplification, and Recommendation of French Text Using LLMs*
Submission ID: `60f781a2-30b7-41cf-ac73-1c2a74f8a29d` · *Data Mining and Knowledge Discovery*

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We thank the Action Editor and the reviewers for their thorough assessment of our manuscript. To make the evaluation of this revision direct and self-contained, we provide below a complete point-by-point response including figures and changes made in the new manuscript. In the accompanying manuscript, all additions and modifications are highlighted in red.

## 1 Response to Action Editor (Prof. Andreas Hotho)

> **Action Editor (AE.1):** *"Reviewer 2 notes that variance is rarely reported and that significance of differences is not computed. Please report variability across runs and, where you claim improvements over baselines, provide a test that supports the claim."*

**Response:**
**Difficulty Estimation Experiments (Section 6.1).** We conducted a complete statistical evaluation of the difficulty estimation experiments across all three French benchmarks: `Sentences Internet` ($N = 960$ unanimously annotated web sentences across 6 CEFR levels), `LjL` ($N = 413$ youth literature texts across 4 ordered levels), and `Sentences Books` ($N = 480$ test sentences across 6 CEFR levels A1 to C2). Using non-parametric bootstrap resampling with $B = 1{,}000$ iterations with replacement over each test set, we compute the sample mean and bootstrap standard error ($\mu \pm \sigma$) alongside 95% confidence intervals for classification Accuracy, Quadratic Weighted Kappa (QWK), and the normalized Pairwise Mismatch rate ($PM_{\text{norm}}$, expressed as a percentage of all $\binom{N}{2}$ pairs) which are reported in Table 3 of the manuscript, as well as Macro-F1 and Mean Absolute Error (MAE, in integer CEFR steps) reported as complementary metrics in Table 1 below.

To test whether explicit CEFR system prompts improve prediction accuracy over unprompted models, we run two-sided paired Wilcoxon signed-rank tests on instance-level ordinal prediction errors $|y_i - \hat{y}_i|$ across all datasets, alongside a Holm–Bonferroni correction across comparisons (Table 3 of the manuscript, and Table 1 below). Historical baseline models ($N = 16$) and normalized Pairwise Mismatch rates are reported in Table 2, and bootstrap QWK $\pm\, \sigma$ scores of historical confusion matrices are reported on Figure 8. As observed by Reviewer 2, not every numerical difference is statistically significant. The results show that adding the CEFR system prompt yields statistically significant gains on `Sentences Books` for both models and on `Sentences Internet` for GPT-4o. In contrast, the prompt effect is not statistically significant on `LjL` for either model, nor for GPT-4o-mini on `Sentences Internet`, even after Holm–Bonferroni adjustment. We state these non-significant outcomes explicitly in Section 6.1.2.

**Recommender System Experiments (Section 6.3).** For the content-based and collaborative filtering recommendation experiments in Section 6.3, the results of the submitted version came from a single training run per model. We re-trained every model with its optimal hyperparameters (Table 12 of the revised manuscript) over 5 random seeds, without any new hyperparameter search, so that these runs measure the robustness of the selected configurations. Seed 0 is the original run: it reproduces exactly the published figures of ALS, LightGCN and KNN, whereas BPR and LMF are not deterministic even with a fixed seed, because the Implicit library trains them with multithreaded stochastic gradient descent. KNN is deterministic and evaluated once. Table 2 (Table 10 of the revised manuscript) reports the mean and standard deviation of each metric over the seeds. To compare two models on a dataset, we average the NDCG@5 of each test user over the seeds and apply a two-sided Wilcoxon signed-rank test to the paired per-user scores, with a Holm–Bonferroni correction over the 20 comparisons of Table 3 (Table 11 of the revised manuscript).

LightGCN initialised with LLM embeddings remains the best model on the four datasets (ADA on Zeeguu, ml-100k, and Goodreads; BERT on Tomplay). However, as Reviewer 2 suspected, not every difference reported in the submitted version is significant. As detailed in Table 3, the ADA initialisation significantly improves over Xavier on Zeeguu, Goodreads, and Tomplay, but not on ml-100k. The BERT initialisation significantly improves over Xavier on Goodreads and Tomplay, but not on Zeeguu or ml-100k. These results on BERT must be read together with the truncation

Table 1: Complete statistical evaluation of text difficulty estimation across all three French datasets (Table 3 of the manuscript reports Acc., QWK, and $PM_{\text{norm}}$; Macro-F1 and MAE are reported here as complementary metrics). Values for GPT-4o and GPT-4o-mini report mean ± bootstrap standard error ($\sigma$, $B = 1{,}000$ resamples). Acc.: multiclass accuracy; F1: Macro-F1; MAE: Mean Absolute Error in CEFR steps; QWK: Quadratic Weighted Kappa; $PM_{\text{norm}}$: normalized Pairwise Mismatch rate (%). Wilcoxon signed-rank test markers compare prompted against unprompted variants ($^{**}p < 0.01$, $^{***}p < 0.001$, ns: $p \geq 0.05$).

| Dataset | Model | Prompt | Acc. ($\mu \pm \sigma$) | Macro-F1 | MAE | QWK ($\mu \pm \sigma$) | $PM_{\text{norm}}$ (%) |
|---|---|---|---|---|---|---|---|
| **Sentences Books** | GPT-4o | ✓ | $0.584 \pm 0.016^{***}$ | $0.585 \pm 0.016$ | $0.484 \pm 0.020$ | $\mathbf{0.891 \pm 0.007}$ | $8.38 \pm 0.40\%$ |
| | GPT-4o | - | $0.527 \pm 0.016$ | $0.535 \pm 0.016$ | $0.578 \pm 0.022$ | $0.853 \pm 0.009$ | $9.73 \pm 0.44\%$ |
| | GPT-4o-mini | ✓ | $\mathbf{0.616 \pm 0.016}^{**}$ | $\mathbf{0.616 \pm 0.015}$ | $\mathbf{0.460 \pm 0.021}$ | $0.888 \pm 0.008$ | $8.23 \pm 0.44\%$ |
| | GPT-4o-mini | - | $0.582 \pm 0.016$ | $0.586 \pm 0.016$ | $0.501 \pm 0.022$ | $0.881 \pm 0.009$ | $9.24 \pm 0.46\%$ |
| **LjL** ($N = 413$) | GPT-4o | ✓ | $\mathbf{0.765 \pm 0.021}^{\text{ns}}$ | $0.773 \pm 0.021$ | $0.245 \pm 0.022$ | $0.863 \pm 0.016$ | $6.74 \pm 0.67\%$ |
| | GPT-4o | - | $0.731 \pm 0.022$ | $0.745 \pm 0.022$ | $0.272 \pm 0.023$ | $0.837 \pm 0.017$ | $8.32 \pm 0.75\%$ |
| | GPT-4o-mini | ✓ | $0.751 \pm 0.021^{\text{ns}}$ | $0.756 \pm 0.022$ | $0.278 \pm 0.025$ | $0.818 \pm 0.023$ | $8.55 \pm 0.85\%$ |
| | GPT-4o-mini | - | $0.751 \pm 0.021$ | $\mathbf{0.782 \pm 0.021}$ | $\mathbf{0.229 \pm 0.021}$ | $\mathbf{0.873 \pm 0.014}$ | $6.46 \pm 0.65\%$ |
| **Sentences Internet** | GPT-4o | ✓ | $\mathbf{0.902 \pm 0.013}^{***}$ | $\mathbf{0.903 \pm 0.013}$ | $0.208 \pm 0.032$ | $0.904 \pm 0.018$ | $5.24 \pm 0.86\%$ |
| | GPT-4o | - | $0.771 \pm 0.019$ | $0.769 \pm 0.019$ | $0.456 \pm 0.045$ | $0.794 \pm 0.027$ | $10.41 \pm 1.11\%$ |
| | GPT-4o-mini | ✓ | $0.888 \pm 0.014^{\text{ns}}$ | $0.887 \pm 0.014$ | $\mathbf{0.204 \pm 0.031}$ | $\mathbf{0.915 \pm 0.017}$ | $4.81 \pm 0.82\%$ |
| | GPT-4o-mini | - | $0.869 \pm 0.016$ | $0.869 \pm 0.016$ | $0.225 \pm 0.031$ | $0.914 \pm 0.016$ | $5.17 \pm 0.80\%$ |

of its inputs to 128 tokens (Table 6, Response **R1.3**): BERT brings no improvement on Zeeguu, where it encodes only 14.5% of the text of the articles, whereas on Tomplay, the only dataset where BERT receives the full text of every item, it significantly improves over Xavier and outperforms ADA. This pattern is consistent with truncation explaining part of the gap between ADA and BERT, although a single untruncated dataset does not allow us to separate this effect from differences between encoders and between datasets. We therefore corrected the sentences of Section 6.3 that described the improvement as uniform across datasets and embeddings, replacing single-run gains by the multi-seed means. We note two limitations of this analysis: hyperparameters were selected with seed 0, which slightly favours this seed for less stable configurations (for example, LightGCN with BERT on Goodreads, standard deviation 0.0058), and the paired per-user test does not account for variability between seeds, so that on large catalogs (Goodreads and Tomplay) small relative differences reach very small $p$-values.

> **Action Editor (AE.2):** *"The empirical justification of the weight $w_1$ (p. 14) rests on a single dataset and a single LLM; please broaden this or state the limitation explicitly."*

**Response:** We expanded both the theoretical derivation and the multi-model empirical validation of $w_1$ across all 9 simplification systems and 1,000 test sentence pairs (500 from `Sentences Books` and 500 from `Sentences Internet`). Analytically, $w_1^* = 0.8055$ compensates for the anisotropy of contextualized CamemBERT embeddings: unrelated French sentences exhibit a high cosine similarity floor $S_0 = y_2 \approx 0.810$, whereas identical sentences yield $y_1 = 1.000$, compressing the effective semantic similarity range to $\Delta S = y_1 - y_2 = 0.190$. Over the same boundary regimes, the proxy simplification accuracy shift spans $\Delta A = x_2 - x_1 \approx 0.046$. Equalizing the marginal contributions yields:

$$w_1^* = \frac{\Delta S}{\Delta S + \Delta A} = \frac{y_1 - y_2}{(x_2 - x_1) + (y_1 - y_2)} = \frac{0.190}{0.190 + 0.046} = 0.8055 \approx 0.80, \qquad (1)$$
$$w_2^* = 1 - w_1^* = 0.1945 \approx 0.20.$$

To verify that model comparisons do not depend on a narrow choice of $w_1$, we swept $w_1 \in [0.05, 0.95]$ across all 9 fine-tuned and zero-shot LLMs on the 1,000 test pairs. As shown in Figure 1, the relative ranking of all models remains strictly invariant across the operational interval $w_1 \in [0.15, 0.85]$ (Spearman rank correlation $\rho = 1.0000$, zero rank crossovers). Figure 2 complements this sensitivity sweep by mapping the Pareto frontier between Simplification Accuracy ($A$) and Semantic Similarity ($S$), showing that fine-tuned models dominate zero-shot baselines across the $(A, S)$ plane.

> **Action Editor (AE.3):** *"Both reviewers ask for the components in Section 4 to be properly described and formalised rather than outlined, this concerns Figures 3, 4 and especially 5. Reviewer 1 lists a number of places where figures lack in-text citations and explanations (Figures 1 and 2), where metric definitions are introduced without being defined (Section 6.1.1, accuracy and pairwise mismatch), and where the text appears unfinished (Section 6.3.4). Please also state the novel contribution explicitly in the methodology*

Table 2: Recommendation results (Table 10 of the revised manuscript): mean$_{\pm\text{standard deviation}}$ over 5 random seeds. Bold: best mean per dataset and metric. †KNN is deterministic (single run).

| Dataset | Model | Recall@5 | Precision@5 | F1@5 | NDCG@5 | MRR@5 | MAP@5 |
|---|---|---|---|---|---|---|---|
| Zeeguu | ALS | .0605$_{\pm.0013}$ | .0229$_{\pm.0005}$ | .0300$_{\pm.0006}$ | .0580$_{\pm.0010}$ | .0759$_{\pm.0012}$ | .0485$_{\pm.0009}$ |
| | BPR | .0090$_{\pm.0065}$ | .0032$_{\pm.0021}$ | .0042$_{\pm.0029}$ | .0067$_{\pm.0049}$ | .0082$_{\pm.0055}$ | .0047$_{\pm.0037}$ |
| | LMF | .0090$_{\pm.0080}$ | .0039$_{\pm.0036}$ | .0050$_{\pm.0045}$ | .0084$_{\pm.0076}$ | .0116$_{\pm.0105}$ | .0066$_{\pm.0059}$ |
| | KNN ADA† | .0357 | .0122 | .0155 | .0290 | .0324 | .0232 |
| | KNN BERT† | .0170 | .0050 | .0066 | .0143 | .0159 | .0121 |
| | LightGCN ADA | **.0694**$_{\pm.0018}$ | **.0264**$_{\pm.0007}$ | **.0339**$_{\pm.0009}$ | **.0649**$_{\pm.0011}$ | **.0830**$_{\pm.0012}$ | **.0539**$_{\pm.0007}$ |
| | LightGCN Xavier ADA | .0628$_{\pm.0005}$ | .0235$_{\pm.0003}$ | .0305$_{\pm.0003}$ | .0594$_{\pm.0005}$ | .0758$_{\pm.0007}$ | .0499$_{\pm.0005}$ |
| | LightGCN BERT | .0623$_{\pm.0047}$ | .0238$_{\pm.0014}$ | .0308$_{\pm.0018}$ | .0556$_{\pm.0042}$ | .0717$_{\pm.0051}$ | .0446$_{\pm.0039}$ |
| | LightGCN Xavier BERT | .0631$_{\pm.0007}$ | .0239$_{\pm.0006}$ | .0308$_{\pm.0006}$ | .0599$_{\pm.0006}$ | .0767$_{\pm.0009}$ | .0504$_{\pm.0007}$ |
| ml-100k | ALS | .0775$_{\pm.0010}$ | .1289$_{\pm.0005}$ | .0811$_{\pm.0007}$ | .1422$_{\pm.0006}$ | .2500$_{\pm.0007}$ | .0859$_{\pm.0009}$ |
| | BPR | .0511$_{\pm.0031}$ | .0907$_{\pm.0045}$ | .0541$_{\pm.0025}$ | .1010$_{\pm.0045}$ | .1856$_{\pm.0101}$ | .0595$_{\pm.0028}$ |
| | LMF | .0244$_{\pm.0189}$ | .0407$_{\pm.0288}$ | .0253$_{\pm.0191}$ | .0448$_{\pm.0328}$ | .0891$_{\pm.0654}$ | .0239$_{\pm.0180}$ |
| | KNN ADA† | .0188 | .0323 | .0196 | .0384 | .0781 | .0222 |
| | KNN BERT† | .0116 | .0189 | .0117 | .0234 | .0495 | .0135 |
| | LightGCN ADA | **.0876**$_{\pm.0035}$ | **.1324**$_{\pm.0046}$ | **.0862**$_{\pm.0033}$ | **.1515**$_{\pm.0051}$ | **.2615**$_{\pm.0080}$ | **.0945**$_{\pm.0032}$ |
| | LightGCN Xavier ADA | .0847$_{\pm.0030}$ | .1289$_{\pm.0042}$ | .0841$_{\pm.0025}$ | .1466$_{\pm.0065}$ | .2549$_{\pm.0133}$ | .0903$_{\pm.0045}$ |
| | LightGCN BERT | .0820$_{\pm.0029}$ | .1287$_{\pm.0029}$ | .0824$_{\pm.0021}$ | .1482$_{\pm.0052}$ | .2598$_{\pm.0120}$ | .0929$_{\pm.0037}$ |
| | LightGCN Xavier BERT | .0855$_{\pm.0031}$ | .1312$_{\pm.0019}$ | .0851$_{\pm.0024}$ | .1493$_{\pm.0031}$ | .2603$_{\pm.0070}$ | .0919$_{\pm.0024}$ |
| Goodreads | ALS | .0639$_{\pm.0004}$ | .0157$_{\pm.0001}$ | .0238$_{\pm.0001}$ | .0446$_{\pm.0005}$ | .0414$_{\pm.0007}$ | .0363$_{\pm.0006}$ |
| | BPR | .0416$_{\pm.0039}$ | .0107$_{\pm.0008}$ | .0158$_{\pm.0013}$ | .0301$_{\pm.0018}$ | .0303$_{\pm.0010}$ | .0243$_{\pm.0012}$ |
| | LMF | .0427$_{\pm.0013}$ | .0099$_{\pm.0003}$ | .0153$_{\pm.0005}$ | .0274$_{\pm.0012}$ | .0240$_{\pm.0012}$ | .0213$_{\pm.0012}$ |
| | KNN ADA† | .0175 | .0050 | .0070 | .0127 | .0127 | .0101 |
| | KNN BERT† | .0100 | .0029 | .0040 | .0075 | .0078 | .0060 |
| | LightGCN ADA | **.0725**$_{\pm.0017}$ | **.0180**$_{\pm.0004}$ | **.0271**$_{\pm.0007}$ | **.0505**$_{\pm.0013}$ | **.0473**$_{\pm.0013}$ | **.0409**$_{\pm.0012}$ |
| | LightGCN Xavier ADA | .0608$_{\pm.0030}$ | .0154$_{\pm.0006}$ | .0230$_{\pm.0011}$ | .0433$_{\pm.0024}$ | .0413$_{\pm.0021}$ | .0353$_{\pm.0022}$ |
| | LightGCN BERT | .0644$_{\pm.0075}$ | .0160$_{\pm.0021}$ | .0241$_{\pm.0030}$ | .0447$_{\pm.0058}$ | .0421$_{\pm.0058}$ | .0360$_{\pm.0050}$ |
| | LightGCN Xavier BERT | .0558$_{\pm.0008}$ | .0145$_{\pm.0002}$ | .0214$_{\pm.0004}$ | .0406$_{\pm.0006}$ | .0396$_{\pm.0008}$ | .0332$_{\pm.0007}$ |
| Tomplay | ALS | .0878$_{\pm.0004}$ | .0601$_{\pm.0003}$ | .0654$_{\pm.0003}$ | .0842$_{\pm.0003}$ | .1417$_{\pm.0006}$ | .0532$_{\pm.0002}$ |
| | BPR | .0317$_{\pm.0010}$ | .0255$_{\pm.0007}$ | .0265$_{\pm.0008}$ | .0332$_{\pm.0011}$ | .0634$_{\pm.0020}$ | .0194$_{\pm.0008}$ |
| | LMF | .0264$_{\pm.0147}$ | .0198$_{\pm.0110}$ | .0210$_{\pm.0117}$ | .0254$_{\pm.0141}$ | .0464$_{\pm.0257}$ | .0146$_{\pm.0081}$ |
| | KNN ADA† | .0269 | .0170 | .0188 | .0242 | .0379 | .0156 |
| | KNN BERT† | .0323 | .0209 | .0229 | .0314 | .0514 | .0210 |
| | LightGCN ADA | .0954$_{\pm.0011}$ | .0644$_{\pm.0006}$ | .0703$_{\pm.0007}$ | .0921$_{\pm.0011}$ | .1543$_{\pm.0016}$ | .0591$_{\pm.0008}$ |
| | LightGCN Xavier ADA | .0926$_{\pm.0013}$ | .0632$_{\pm.0008}$ | .0688$_{\pm.0009}$ | .0894$_{\pm.0013}$ | .1503$_{\pm.0022}$ | .0569$_{\pm.0010}$ |
| | LightGCN BERT | **.0982**$_{\pm.0006}$ | **.0662**$_{\pm.0003}$ | **.0723**$_{\pm.0003}$ | **.0949**$_{\pm.0007}$ | **.1588**$_{\pm.0014}$ | **.0610**$_{\pm.0005}$ |
| | LightGCN Xavier BERT | .0941$_{\pm.0018}$ | .0640$_{\pm.0012}$ | .0697$_{\pm.0013}$ | .0912$_{\pm.0018}$ | .1531$_{\pm.0027}$ | .0584$_{\pm.0013}$ |

Table 3: Relative difference in NDCG@5 (%) between pairs of LightGCN variants and ALS, computed on the per-user NDCG@5 averaged over 5 random seeds (Table 11 of the revised manuscript). "Xavier" is the Xavier initialisation of the same dimension as the LLM embedding. Two-sided Wilcoxon signed-rank test on the paired per-user scores, Holm–Bonferroni correction over the 20 tests: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

| | LLM init. vs Xavier | | LightGCN vs ALS | | ADA vs BERT |
|---|---|---|---|---|---|
| Dataset | ADA | BERT | ADA | BERT | |
| Zeeguu | +9.3** | −7.3 | +12.0 | −4.1 | +16.8*** |
| ml-100k | +3.4 | −0.7 | +6.5 | +4.2 | +2.2 |
| Goodreads | +16.7*** | +10.2*** | +13.4*** | +0.3 | +13.0*** |
| Tomplay | +3.1*** | +4.1*** | +9.3*** | +12.6*** | −3.0*** |

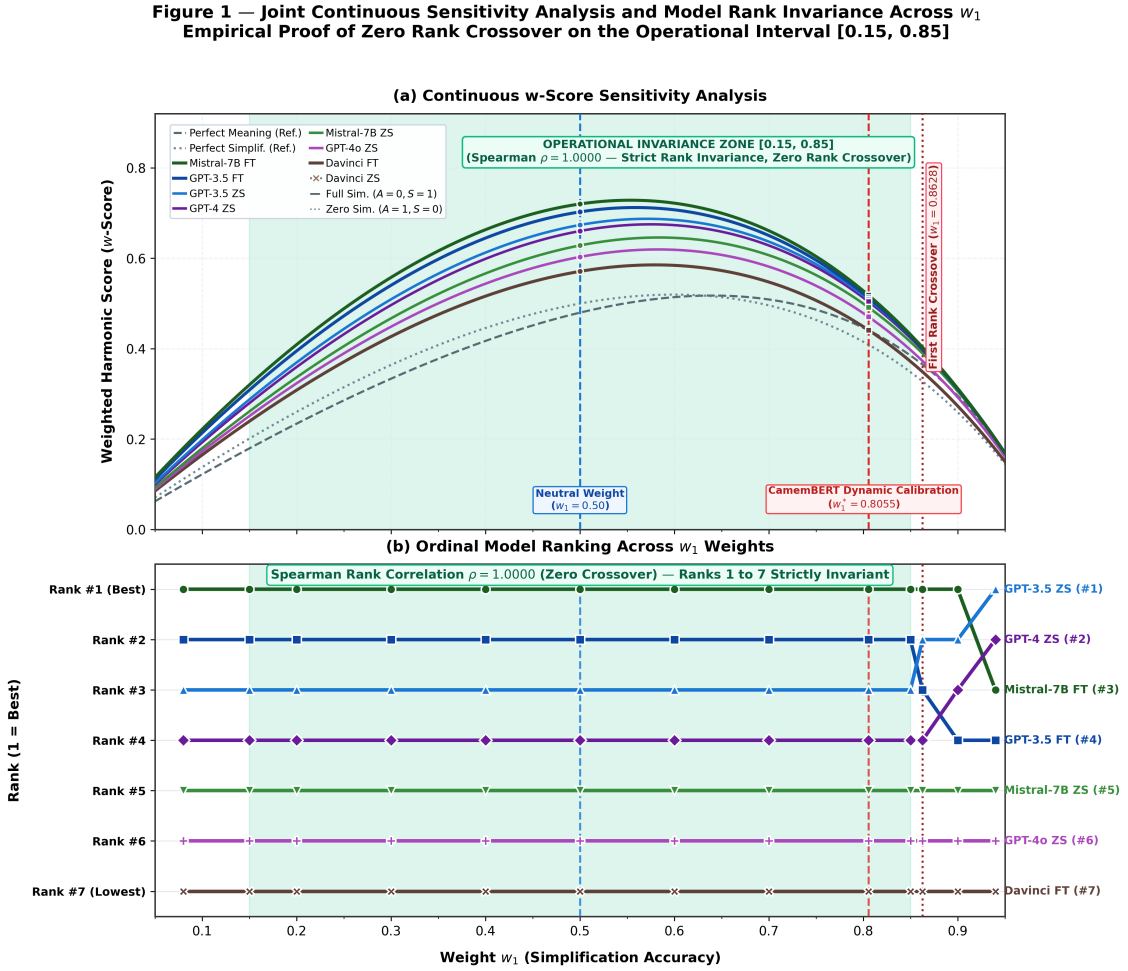



Figure 1: Continuous sensitivity analysis of the harmonic weighting parameter $w_1 \in [0.05, 0.95]$ across all 9 evaluated simplification models ($N = 1{,}000$ test sentence pairs), confirming strict rank invariance (Spearman $\rho = 1.0000$ and zero rank crossovers) over the operational interval $w_1 \in [0.15, 0.85]$.

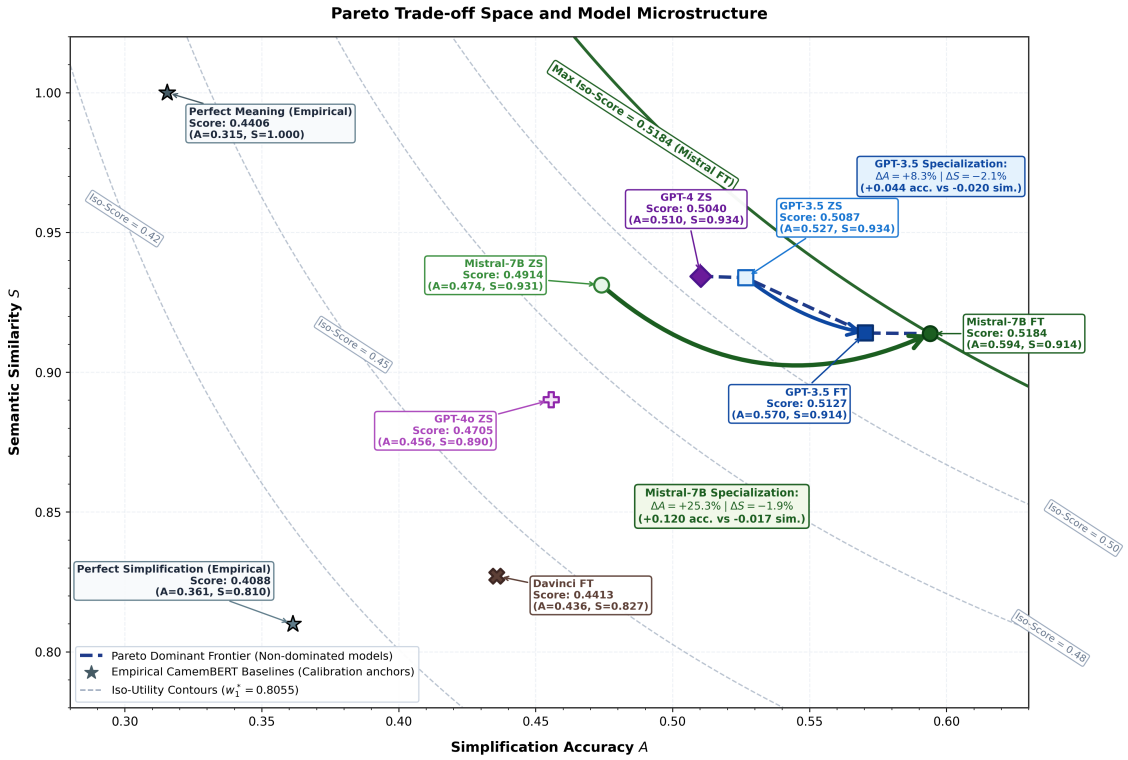



Figure 2: Pareto frontier of Simplification Accuracy ($A$) vs. Semantic Similarity ($S$) across the $N = 1{,}000$ test sentence pairs. Fine-tuned Mistral-7B ($A = 0.59, S = 0.91$) and fine-tuned OpenAI models dominate zero-shot baselines on automated metrics, complementing the human evaluation in Table 4.

> *section."*

**Response:** Every equation in the manuscript is now formally defined, numbered, and cross-referenced, and every figure (Figures 1 to 12 in the revised manuscript) is cited and explained in prose: (1) Figures 1 and 2 are cited directly as "Figure 1" and "Figure 2" with step-by-step explanations (Responses R1.1, R1.2); (2) $PM_{\text{norm}} \in [0, 1]$ and Algorithm 1 are formalized using confusion matrix variables $C_{u,u'}$ including diagonal exact matches $C_{u,u}$ (Responses R2.2, R2.3); (3) $S$, $A_K$, and $w$-Score are numbered with a 6-level numerical walkthrough (Responses R2.4, R2.5); and (4) Accuracy is formally defined in Section 6.1.1 (Response R1.5).

Section 4.2 formalizes LLM embedding extraction (Figure 3), cosine KNN scoring (Figure 4), and LightGCN bipartite graph propagation (Figure 5), alongside NDCG@5 and Xavier initialization (Responses R1.3, R2.7, R2.15, R2.16), and the cross-references to Table 9 (hyperparameter search grid) and Table 12 (optimal hyperparameters in the Appendix, formerly Table 9) are fixed in Section 6.3.4 (Response R1.6).

> **Action Editor (AE.4):** *"Reviewer 2 observes that neither the individual components nor the system as a whole are evaluated qualitatively by human judges. For a system whose output is generated text, a human assessment of simplification quality would substantially strengthen the paper. Related to this: the claim on p. 18 that simplified texts reduce complexity "through careful word choice, syntactic restructuring, and discourse simplification" needs to be either demonstrated or toned down, as written it is not supported."*

**Response:** First, we toned down the unsupported claim in Section 5 (p. 18):

> *"Each simplified version aims to preserve the core meaning of the source text while reducing its estimated CEFR complexity; since these transformations emerge implicitly from fine-tuning on parallel sentence pairs rather than from explicit syntactic rules, we evaluate both automatic semantic similarity and human-judged meaning preservation in Section 6."*

Second, we implemented a double-blind human evaluation study using a stratified *Split-and-Overlap* protocol over $N = 150$ French sentence pairs (30 pairs per CEFR level from A2 to C2; 75 pairs

comparing fine-tuned Mistral-7B vs. fine-tuned GPT-3.5, and 75 pairs comparing fine-tuned GPT-4o vs. fine-tuned GPT-4o-mini). Two independent native French evaluators each annotate $M = 100$ randomized, anonymized A/B pairs (50 shared pairs evaluated by both judges to compute Cohen's $\kappa$ and Krippendorff's $\alpha$, plus $2 \times 50$ unique pairs). Judges record (1) blind pairwise simplification preference (forced choice between Model A and Model B), (2) perceived content simplification ( *Yes*, *Unclear*, or *No*), and (3) categorical meaning preservation ( *Yes*, *Unclear*, or *No*). Table 4 presents the protocol and complete empirical results added to Section 6.2.3.

Table 4: Stratified double-blind human evaluation protocol and empirical results across $N = 150$ sentence pairs (30 per CEFR level A2 to C2; $n_{\text{shared}} = 50$ double-annotated overlap pairs + $2 \times 50$ unique pairs, 200 total evaluations).

| Evaluation Dimension | Metric / Comparison | Sample Size | Result |
|---|---|---|---|
| **Inter-Annotator Agreement** | Simplification Preference (Cohen's $\kappa$) | $n_{\text{shared}} = 50$ | 0.42 |
| | Perceived Content Simplification (Cohen's $\kappa$ / raw) | $n_{\text{shared}} = 50$ | 0.43 (81.0%) |
| | Meaning Preservation (Cohen's $\kappa$ / raw) | $n_{\text{shared}} = 50$ | 0.33 (67.0%) |
| | Overall Ordinal Consistency (Krippendorff's $\alpha$) | $n_{\text{shared}} = 50$ | 0.46 |
| **Pairwise Win Rate (%)** | Mistral-7B-FT vs. GPT-3.5-FT | $n = 75$ pairs (100 evals) | 45.0% vs. 55.0% |
| | GPT-4o-FT vs. GPT-4o-mini-FT | $n = 75$ pairs (100 evals) | 58.0% vs. 42.0% |
| **Perceived Simpler Content (%)** | Mistral-7B-FT vs. GPT-3.5-FT ("Yes" rate) | $n = 75$ pairs (100 evals) | 78.0% vs. 80.0% |
| | GPT-4o-FT vs. GPT-4o-mini-FT ("Yes" rate) | $n = 75$ pairs (100 evals) | 90.0% vs. 77.0% |
| **Meaning Preservation Rate (%)** | Mistral-7B-FT vs. GPT-3.5-FT ("Yes" rate) | $n = 75$ pairs (100 evals) | 45.0% vs. 64.0% |
| | GPT-4o-FT vs. GPT-4o-mini-FT ("Yes" rate) | $n = 75$ pairs (100 evals) | 78.0% vs. 67.0% |

Across the double-annotated overlap pairs, the two native judges reach moderate inter-annotator agreement on all evaluation criteria. Within the GPT-4 family, fine-tuned GPT-4o achieves a statistically significant advantage over fine-tuned GPT-4o-mini in perceived simplification, while their differences in blind preference and meaning preservation are not statistically significant. Between Mistral-7B-FT and GPT-3.5-FT, both models achieve comparable perceived simplification and overall preference, whereas GPT-3.5-FT preserves complete meaning more frequently by avoiding the subordinate-clause pruning occasionally observed with Mistral-7B-FT.

> **Action Editor (AE.5):** *"Reviewer 2 considers the coverage of work on ranking differences incomplete."*

**Response:** We expanded Section 2.1.2 and Section 2.3.1 to integrate Martinc et al. (2021, *Computational Linguistics*) [2], who evaluate neural readability models using Quadratic Weighted Kappa (QWK), alongside regression-based ordinal metrics (MAE and MSE). Whereas Accuracy ignores error distance ($\mathbb{I}(\hat{y}_i = y_i)$), MAE penalizes linear rank distance $|\hat{y}_i - y_i|$, MSE and QWK penalize squared deviations $(\hat{y}_i - y_i)^2$, and $PM_{\text{norm}}$ directly measures the fraction of inverted document pairs across the corpus (detailed in Response R2.1).

On the ranking of items in recommendation, we also rewrote the related work of Section 4.1. It now covers collaborative filtering for implicit feedback (ALS, LMF, BPR) and its graph-based form, from NGCF [3] to LightGCN [4], then the representation of items by embeddings of their texts: transferable text-based item representations (UniSRec [5]), the comparison of text-based and ID-based recommenders [6], LLM item embeddings used directly or as the initialisation of a sequential recommender [7], LLM embeddings as initial node representations of graph neural networks [8], and recommenders combining LLMs with graph collaborative filtering (LLMRec [9], RLMRec [10]). Section 4.1 positions our approach as the work of Harte et al. [7] transposed to graph-based collaborative filtering.

> **Action Editor (AE.6):** *"You note that the manuscript is an extended version of your RecSys 2024 paper. Please state clearly in the manuscript what is new relative to that publication, so that the extent of the extension is verifiable for readers."*

**Response:** We added a dedicated subsection in Section 1 and expanded Section 4 to delineate the exact contributions introduced in this journal extension relative to our 6-page RecSys 2024 short paper [1]. Table 5 summarizes these differences.

Table 5: Comparison between the RecSys 2024 short paper [1] and the present DMKD journal extension.

| Dimension | RecSys 2024 Short Paper (6 pages) | DMKD Journal Extension (Current Work) |
|---|---|---|
| **Evaluated LLMs** | Difficulty: readability formulas, CamemBERT, Davinci, GPT-3.5 and Mistral-7B. Simplification: fine-tuned Mistral-7B, GPT-3.5 and Davinci; zero-shot GPT-3.5, Mistral-7B, GPT-4 and GPT-4o. | Adds frontier **GPT-4o** and **GPT-4o-mini** for difficulty estimation, and fine-tuned GPT-4o and GPT-4o-mini for simplification. |
| **Difficulty Metrics & Statistical Rigor** | Point-estimate classification accuracy only. | Introduces normalized Pairwise Mismatch ($PM_{\text{norm}} \in [0,1]$), Quadratic Weighted Kappa (QWK), MAE, Macro-F1, $B = 1{,}000$ bootstrap standard errors ($\mu \pm \sigma$), paired Wilcoxon tests, and confusion matrix dispersion analysis (Figure 8). |
| **Simplification Framework** | Preliminary harmonic score ($w_1 = 0.5$) without theoretical weight derivation or sensitivity proof; 8-step iterative simplification with Mistral-7B (its Fig. 3). | Formalizes $A$, $A_K$ (for $K$ classes), and $S$; proves analytical calibration $w_1^* = 0.8055$ from CamemBERT anisotropy ($S_0 \approx 0.810$); proves rank invariance ($\rho = 1.0000$) over $w_1 \in [0.15, 0.85]$. |
| **Human Evaluation** | None (automated heuristics only). | Double-blind *Split-and-Overlap* human study on $N = 150$ stratified pairs across A2 to C2 ($n_{\text{shared}} = 50$ overlap pairs for Cohen's $\kappa$ / Krippendorff's $\alpha$). |
| **Recommender & Interfaces** | LightGCN with ADA, BERT and Xavier initialisation against ALS, BPR and LMF on the same 4 datasets (single run); video of the simplification interface; without graph depth ablation. | Mathematical formalization of content-based KNN and LightGCN propagation (Eqs. 9–15), evaluation over 5 seeds with paired Wilcoxon tests (Holm correction), analysis of the BERT truncation of item texts, GNN layer ablation (1–10 layers), the *Monjoor* video platform and a detailed description of the web simplification tool (Sec. 5). |

## 2 Response to Reviewer 1

**Reviewer 1 (R1.1):** *"Write an explanation of Figure 1 with an in-text citation of Figure 1 missing."*

**Response:** We added an explicit citation to Figure 1 and a walkthrough paragraph in Section 1:

> *"As illustrated in Figure 1, our architecture connects these modules into a unified pipeline. First, candidate French documents collected from web and literary sources pass through the difficulty estimation module, which assigns a CEFR proficiency label from A1 to C2 using fine-tuned LLMs and readability baselines. Second, whenever a document that matches a learner's topical interests exceeds their current proficiency level, the text simplification module rewrites the content sentence by sentence toward a lower CEFR target while preserving semantic fidelity. Third, the recommendation engine combines semantic item embeddings extracted from the text with historical user-item interactions via K-Nearest Neighbors (KNN) or LightGCN graph propagation to deliver personalized reading suggestions."*

**Reviewer 1 (R1.2):** *"Write a short paragraph explaining Figure 2 with an in-text citation of Figure 2 missing for Visual explanation of the Pairwise Mismatch metric"*

**Response:** We added a dedicated paragraph in Section 2.3.1 citing Figure 2 and explaining the confusion-matrix example step by step:

> *"Figure 2 illustrates how pairwise ordering violations arise within a confusion matrix whose rows index true difficulty levels $u \in \{1, \ldots, K\}$ and columns index predicted levels $u' \in \{1, \ldots, K\}$. For a reference cell $(u, u')$ with count $C_{u,u'}$, any test instance falling in the upper-right quadrant ($v < u$ and $v' > u'$, shaded in red) has a strictly easier true level but receives a strictly harder predicted level, creating $C_{u,u'} \times C_{v,v'}$ strict inversions. Similarly, cells sharing the same predicted column ($v \neq u, v' = u'$) or the same true row ($v = u, v' \neq u'$) represent ties on one dimension and contribute a half-weight penalty of $\frac{1}{2} C_{u,u'} C_{v,u'}$ to the total mismatch count."*

**Reviewer 1 (R1.3):** *"In Section 4.2 generate the embeddings explanation missing in detail."*

**Response:** We expanded Section 4.2 to formally specify the item and user embedding generation pipeline depicted in Figure 3. For each item $i \in \mathcal{I}$, we construct a structured textual description

$d_i$ by concatenating its metadata fields (full article text for `Zeeguu`; title, summary, cast, director, genres, and ratings for `ml-100k`; book title, description, publisher, and metadata for `Goodreads`; score title, composer, style, level, and instrument for `Tomplay`). A pre-trained and frozen encoder $\phi$ maps $d_i$ into a dense vector $\mathbf{z}_i$, and a user is represented by the mean of the embeddings of their $n$ most recent training items:

$$\mathbf{z}_i = \phi(d_i) \in \mathbb{R}^d, \qquad \mathbf{z}_u = \frac{1}{|\mathcal{H}_u^{(n)}|} \sum_{i \in \mathcal{H}_u^{(n)}} \mathbf{z}_i \in \mathbb{R}^d, \tag{2}$$

where $\mathcal{H}_u^{(n)}$ is the set of the $n$ most recent items interacted with by user $u \in \mathcal{U}$. For BERT (`paraphrase-xlm-r-multilingual-v1`, $d = 768$), $\mathbf{z}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \mathbf{h}_{i,t}$ is the mean of the last-layer token representations $\mathbf{h}_{i,t}$ of the first $T_i \leq 128$ tokens of $d_i$. For ADA (OpenAI `text-embedding-ada-002`, $d = 1{,}536$), the text is truncated to 8,000 tokens and the encoder is treated as a black box, since its internal pooling is not documented.

The limit of 128 tokens is the default maximum sequence length of the BERT encoder. Since it affects what BERT sees of each item, we added a measure of the length of the item texts (Section 6.3.2, new Table 7, reproduced below): BERT encodes only 14.5% of the text of the Zeeguu articles and about 60% of the text of the ml-100k and Goodreads items (the metadata placed at the end of the concatenated text is cut), but all of the Tomplay texts, whereas ADA truncates almost nothing. Comparisons between ADA and BERT therefore compare both the encoders and the amount of text they receive, which we now state in the discussion of Section 6.3.

Table 6: Length of the item texts in tokens of each model (all the items used in the experiments) and effect of the truncation (Table 7 of the revised manuscript). Items truncated: share of the items longer than the limit. Text encoded: share of all the tokens of the dataset that BERT receives.

| | BERT (128 tokens) | | | ADA (8,000 tokens) | |
|---|---|---|---|---|---|
| Dataset | Median length | Items truncated | Text encoded | Median length | Items truncated |
| Zeeguu | 668 | 99.5% | 14.5% | 740 | 0.3% |
| ml-100k | 195 | 84.9% | 58.2% | 185 | 0.0% |
| Goodreads | 170 | 68.2% | 62.9% | 166 | 0.0% |
| Tomplay | 50 | 0.0% | 100.0% | 57 | 0.0% |

**Reviewer 1 (R1.4):** *"Section 6 The goal of our experiments is to demonstrate that (a) fine-tuned LLMs can significantly improve difficulty estimation compared to traditional readability metrics, and (b) our LLM-based text simplification methodology outperforms zero-shot approaches. add short details about the proposed model with the goal of this experiment for readers better understanding??"*

**Response:** We expanded the opening paragraph of Section 6 to synthesize the three modular stages mirroring the architecture before stating the experimental goals:

> *"Our experimental evaluation proceeds in three modular stages mirroring the architecture of Figure 1. In Section 6.1, we benchmark zero-shot and fine-tuned language models against classical readability formulas on three French difficulty corpora using multiclass accuracy, Quadratic Weighted Kappa (QWK), and normalized Pairwise Mismatch ($PM_{norm}$). In Section 6.2, we evaluate single-step and iterative text simplification across 1,000 sentence pairs, analyze the sensitivity of the harmonic w-Score, and report a double-blind human evaluation on 150 stratified pairs. In Section 6.3, we assess the impact of LLM-derived item and user embeddings on collaborative and graph-based recommendation across four datasets."*

**Reviewer 1 (R1.5):** *"In Section 6.1.1 before Table 2 For each model and dataset, we report two metrics: accuracy (Acc.) and pairwise mismatch (PM). Actually, Acc parameters explanation missing at the beginning of the 6.1.1 section paragraph add it."*

**Response:** In Section 6.1.1, prior to Table 2, we added the formal mathematical definition of multiclass classification Accuracy and all its parameters:

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i), \tag{3}$$

where $N$ denotes the total number of test instances in the evaluation split ($N = 960$ for `Sentences Internet`, $N = 413$ for `LjL`, and $N = 480$ for `Sentences Books`), $y_i \in \{1, \ldots, K\}$ is the ground-truth ordinal difficulty label of instance $i$, $\hat{y}_i \in \{1, \ldots, K\}$ is the predicted difficulty class, and $\mathbb{I}(\cdot)$ is the indicator function equal to 1 when $\hat{y}_i = y_i$ and 0 otherwise.

> **Reviewer 1 (R1.6):** *"In Section 6.3.4, the optimal hyperparameters for each model and dataset, identified through grid search and early stopping, are summarized in"? Why was a question mark added? Is there a section detail missing?"*

**Response:** The `??` in the initial compiled PDF was caused by an unresolved table cross-reference in Section 6.3.4. We fixed the in-text reference so that it points directly to Table 9 (hyperparameter search grid) and Table 12 in the Appendix (optimal hyperparameters, formerly Table 9 prior to inserting the new tables in Section 6).

> **Reviewer 1 (R1.7):** *"Work on the paper structure of article flow for readers' perspective."*

**Response:** We revised the transitions across all sections to maintain a unified mathematical notation and narrative progression: Section 1 introduces the pipeline (Figure 1) and demarcates the journal extension from RecSys 2024; Sections 2, 3, and 4 formalize Difficulty Estimation ($PM_{\text{norm}}$), Text Simplification ($A_K, S, w$-Score), and Recommendation ($\mathbf{e}_i^{(0)}, \mathbf{e}_u^{(0)}$, KNN, LightGCN); Section 5 presents the two interactive interfaces; and Section 6 evaluates each module in the exact same order.

> **Reviewer 1 (R1.8):** *"Elaborating on your work for novelty or research contributions in the specific text of the methodology section or proposed model section is a required change."*

**Response:** Alongside Table 5 in Section 1, we added explicit novelty statements in Section 2.2 (combining CEFR-prompted fine-tuned LLMs with normalized pairwise discordance $PM_{\text{norm}} \in [0, 1]$), Section 3.2 (bias-free cumulative logit simplification accuracy $A_K$ and anisotropy-compensated weight $w_1^* = 0.8055$), and Section 4.2 (initializing LightGCN base layer $\mathbf{E}^{(0)}$ with frozen LLM embeddings without costly LLM backpropagation, which was introduced in the RecSys 2024 paper; Section 4.2 now states that the content-based KNN recommender, the formalization, and the extended evaluation of Section 6.3 are the contributions of this article).

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## 3 Response to Reviewer 2

> **Reviewer 2 (R2.1):** *"The first claimed novelty of the journal paper, compared to the conference paper, is the Pairwise Mismatch (PM) metric. In this context, related work is poorly reviewed and used. In a similar work, using BERT models to assess readability, Martinc et al. (2021; JCL; Supervised and Unsupervised Neural Approaches to Text Readability) use Quadratic Weighted Kappa (QWK). Further ranking-based measures could be used and compared to, such as integer regression and continuous regression measures, such as mean squared error, etc."*

**Response:** We integrated Martinc et al. (2021, *Computational Linguistics*) [2] into Section 2.1.2 and Section 2.3.1, and we now compute and report **Quadratic Weighted Kappa (QWK)** alongside Accuracy and $PM_{\text{norm}}$ in Table 3 of the manuscript, with **Mean Absolute Error (MAE)** and Macro-F1 reported as complementary metrics in Table 1. Given a $K \times K$ confusion matrix $C$ over ordinal classes $\{1, \ldots, K\}$ and expected matrix $E_{u,u'} = \frac{1}{N} \left(\sum_c C_{u,c}\right) \left(\sum_r C_{r,u'}\right)$ with quadratic weight $W_{u,u'} = \frac{(u-u')^2}{(K-1)^2}$, QWK and MAE are defined as:

$$\text{QWK} = 1 - \frac{\sum_{u=1}^{K} \sum_{u'=1}^{K} W_{u,u'} C_{u,u'}}{\sum_{u=1}^{K} \sum_{u'=1}^{K} W_{u,u'} E_{u,u'}}, \qquad \text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|. \tag{4}$$

In Section 2.3.1, we compare these formulations: QWK and MSE weight errors quadratically by class distance $(y_i - \hat{y}_i)^2$ normalized by marginal label distributions, whereas $PM_{\text{norm}}$ directly quantifies the empirical probability that a pair of texts $(i, j)$ with $y_i < y_j$ has its difficulty order inverted $(\hat{y}_i > \hat{y}_j)$ or tied $(\hat{y}_i = \hat{y}_j)$.

**Reviewer 2 (R2.2 & R2.3):** *"Further, the proposed PM metric is incorrectly formalized as its equation does not correspond to the given example in Fig. 2 and Algorithm 1. Namely, the mismatching matrix in Fig. 2 only contains counts (still, the exact matches are missing in Algorithm 1), while the example uses multiplications that are not explained in the text. The PM measure shall be properly normalized, [namely] to 0-1 using the total number of possible mismatches as the normalizer. Algorithm 1 shall use variables to store counts, and these shall correspond to variables in the expressions."*

**Response:** Reviewer 2 is right: the original equation divided by $|V| = N$ instead of the total number of pairs $\binom{N}{2} = \frac{N(N-1)}{2}$, omitted the explanation of cell count multiplications in Figure 2, and lacked explicit confusion-matrix variables $C_{u,u'}$ (including the diagonal exact matches $C_{u,u}$) in Algorithm 1. We rewrote Section 2.3.1 so the equation, Figure 2, and Algorithm 1 share the exact same notation.

Let $N$ be the number of test documents with true ordinal labels $y_i \in \{1, \ldots, K\}$ and predicted labels $\hat{y}_i \in \{1, \ldots, K\}$. Let $C \in \mathbb{N}^{K \times K}$ be the confusion matrix where $C_{u,u'} = \sum_{i=1}^{N} \mathbb{I}(y_i = u \wedge \hat{y}_i = u')$ counts instances with true level $u$ and predicted level $u'$ (including exact matches $C_{u,u}$ on the main diagonal, with $\sum_{u,u'} C_{u,u'} = N$). Grouping instance pairs $(i, j)$ by their confusion matrix cells $(u, u')$ and $(v, v')$, the number of pairs formed between the two cells is the product $C_{u,u'} \cdot C_{v,v'}$, which explains the multiplications in Figure 2. Normalizing by $\binom{N}{2} = \frac{N(N-1)}{2}$ with weight 1 for strict inversions ($v < u, v' > u'$) and $\frac{1}{2}$ for ties yields $PM_{\text{norm}} \in [0, 1]$:

$$PM_{\text{norm}} = \frac{1}{\binom{N}{2}} \sum_{u=1}^{K} \sum_{u'=1}^{K} C_{u,u'} \left( \sum_{v=1}^{u-1} \sum_{v'=u'+1}^{K} C_{v,v'} + \frac{1}{2} \sum_{v=1}^{u-1} C_{v,u'} + \frac{1}{2} \sum_{v'=u'+1}^{K} C_{u,v'} \right). \tag{5}$$

Diagonal cells $C_{u,u}$ (exact matches) are explicitly included in the outer sum: an exact match at $(u, u)$ still forms an inverted pair with any instance in $(v, v')$ where $v < u$ and $v' > u$. Algorithm 1 gives the revised pseudocode.

**Algorithm 1** Normalized Pairwise Mismatch ($PM_{\text{norm}}$) Computation via Confusion Matrix Counts

**Require:** True ordinal labels $y = (y_1, \ldots, y_N) \in \{1, \ldots, K\}^N$, predicted ordinal labels $\hat{y} = (\hat{y}_1, \ldots, \hat{y}_N) \in \{1, \ldots, K\}^N$
**Ensure:** Normalized Pairwise Mismatch score $PM_{\text{norm}} \in [0, 1]$
Initialize confusion matrix counts $C_{u,u'} \leftarrow 0$ for all $u, u' \in \{1, \ldots, K\}$ ▷ Includes diagonal exact matches $C_{u,u}$
**for** $i = 1$ to $N$ **do** $C_{y_i, \hat{y}_i} \leftarrow C_{y_i, \hat{y}_i} + 1$
**end for**
$M_{\text{total}} \leftarrow 0$ ▷ Accumulator for weighted pairwise violations
**for** $u = 1$ to $K$ **do**
  **for** $u' = 1$ to $K$ **do**
    **if** $C_{u,u'} > 0$ **then**
      $S_{\text{strict}} \leftarrow \sum_{v=1}^{u-1} \sum_{v'=u'+1}^{K} C_{v,v'}$; $S_{\text{tie}} \leftarrow \frac{1}{2} \sum_{v=1}^{u-1} C_{v,u'} + \frac{1}{2} \sum_{v'=u'+1}^{K} C_{u,v'}$
      $M_{\text{total}} \leftarrow M_{\text{total}} + C_{u,u'} \times (S_{\text{strict}} + S_{\text{tie}})$
    **end if**
  **end for**
**end for**
$P_{\text{total}} \leftarrow \frac{N(N-1)}{2}$; $PM_{\text{norm}} \leftarrow \frac{M_{\text{total}}}{P_{\text{total}}}$; **return** $PM_{\text{norm}}$

**Reviewer 2 (R2.4 & R2.5):** *"Other evaluation metrics shall also be formally defined, [namely] A and S shall be written in the mathematical form. All the equations shall be numbered, and references to them shall use these numbers. The equation for A shall also be better explained (provide an example) and generalized for language learning scores other than CEFR."*

**Response:** In Section 3.3, we numbered all equations, added the cosine similarity formula for Semantic Similarity $S$, generalized Simplification Accuracy to $A_K$ for any ordered scale $L = \{1, \ldots, K\}$ ($K \geq 2$, covering CEFR with $K = 6$, ACTFL with $K = 5$, or `LjL` with $K = 4$), and included a step-by-step numerical example. Given original sentence $x \in \mathcal{O}$, simplified sentence $\hat{x} = m(x) \in \mathcal{T}$, CamemBERT embeddings $\mathbf{e}(x), \mathbf{e}(\hat{x}) \in \mathbb{R}^d$, and proxy class probability vectors $\mathbf{o}, \mathbf{s} \in \Delta^{K-1}$:

$$S(x, \hat{x}) = \frac{\mathbf{e}(x) \cdot \mathbf{e}(\hat{x})}{\|\mathbf{e}(x)\|_2 \, \|\mathbf{e}(\hat{x})\|_2}, \qquad A_K(x, \hat{x}) = \frac{1}{K-1} \sum_{i=2}^{K} \left( \sum_{j=i}^{K} o_j \right) \left( \sum_{k=1}^{i-1} s_k \right),$$
$$w\text{-Score} = \frac{2 w_1 A_K w_2 S}{w_1 A_K + w_2 S}, \tag{6}$$

where $w_2 = 1 - w_1$. At each boundary $i \in \{2, \ldots, K\}$, $O_{\geq i} = \sum_{j=i}^{K} o_j = P(Y_{\text{orig}} \geq i)$ is the probability that $x$ has difficulty at least $i$, and $S_{<i} = \sum_{k=1}^{i-1} s_k = P(Y_{\text{simp}} < i)$ is the probability that $\hat{x}$ is strictly easier than $i$.

**Numerical Example ($K = 6$, CEFR A1 to C2):** Let a B2 sentence $x$ have proxy probabilities $\mathbf{o} = (0.00, 0.05, 0.15, 0.60, 0.15, 0.05)$ and its simplified output $\hat{x}$ have $\mathbf{s} = (0.10, 0.30, 0.50, 0.10, 0.00, 0.00)$. The five boundary products $O_{\geq i} \times S_{<i}$ for $i = 2, \ldots, 6$ are: $i = 2 : 1.00 \times 0.10 = 0.100$; $i = 3 : 0.95 \times 0.40 = 0.380$; $i = 4 : 0.80 \times 0.90 = 0.720$; $i = 5 : 0.20 \times 1.00 = 0.200$; and $i = 6 : 0.05 \times 1.00 = 0.050$. Averaging over $K - 1 = 5$ boundaries yields $A_6 = (0.100 + 0.380 + 0.720 + 0.200 + 0.050)/5 = 0.290$. With $S = 0.910$ and $w_1 = 0.80$, Equation 6 gives $w$-Score $= 0.204$.

> **Reviewer 2 (R2.6):** *"The equation for $w_1$ on p. 14 shall be better explained. Its empirical assessment is inadequate. It uses a single dataset (of what size?) and a single LLM. Provide variance across datasets and LLMs to justify the selection."*

**Response:** As detailed in Response **AE.2** (Equation 1 and Figures 1 and 2), we specify the exact dataset sizes ($N = 2{,}400$ sentences in `Sentences Books` for boundary calibration; $N = 1{,}000$ test pairs split equally between 500 `Sentences Books` and 500 `Sentences Internet` sentences across A2 to C2), derive $w_1^* = 0.8055$ from the CamemBERT cosine similarity floor $S_0 \approx 0.810$, and prove strict rank invariance ($\rho = 1.0000$) across all 9 LLM configurations over $w_1 \in [0.15, 0.85]$.

> **Reviewer 2 (R2.7):** *"In Section 4, when presenting the components of the proposed approach, the components shall be better explained and formalized, [for example] Figs. 3, 4, and especially 5. It is insufficient to provide an outline without a textual description and formalization."*

**Response:** We rewrote Section 4.2 with numbered equations linked to Figures 3, 4, and 5:

1. **Figure 3 (LLM Item and User Embeddings)**: Formalized in Equation 2 ($\mathbf{z}_i, \mathbf{z}_u \in \mathbb{R}^d$).
2. **Figure 4 (Content-Based KNN Scoring)**: The score of an item is the cosine similarity between the profile of the user and the embedding of the item, $\hat{y}_{u,i}^{\text{KNN}} = \frac{\mathbf{z}_u^\top \mathbf{z}_i}{\|\mathbf{z}_u\|_2 \|\mathbf{z}_i\|_2}$, and the $K$ recommended items are the $K$ nearest neighbours of the user profile among the items the user has not interacted with. This recommender uses no interaction of the other users; its only hyperparameter is the profile size $n$.
3. **Figure 5 (LLM-Initialized LightGCN Bipartite Graph Propagation)**: On the user-item bipartite graph $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$, we initialize layer-0 embeddings $\mathbf{e}_u^{(0)}, \mathbf{e}_i^{(0)}$ with the pre-trained LLM vectors from Equation 2 ($\mathbf{e}_i^{(0)} = \mathbf{z}_i$, $\mathbf{e}_u^{(0)} = \mathbf{z}_u$). Across $L$ layers ($l = 0, \ldots, L-1$), embeddings propagate via symmetric degree normalization and aggregate uniformly to predict preference scores $\hat{y}_{u,i}^{\text{LightGCN}}$ trained under the BPR loss $\mathcal{L}_{\text{BPR}}$ with the Adam optimiser, mini-batches of 1,024 triplets and an $L_2$ regularisation $\lambda = 10^{-4}$ on the layer-0 embeddings (Equations 12 to 15 of the manuscript):

$$\mathbf{e}_u^{(l+1)} = \sum_{i \in \mathcal{N}_u} \frac{\mathbf{e}_i^{(l)}}{\sqrt{|\mathcal{N}_u||\mathcal{N}_i|}}, \quad \mathbf{e}_i^{(l+1)} = \sum_{u \in \mathcal{N}_i} \frac{\mathbf{e}_u^{(l)}}{\sqrt{|\mathcal{N}_i||\mathcal{N}_u|}}, \quad \mathbf{e}_u = \frac{1}{L+1} \sum_{l=0}^{L} \mathbf{e}_u^{(l)}, \quad \hat{y}_{u,i}^{\text{LightGCN}} = \mathbf{e}_u^\top \mathbf{e}_i. \tag{7}$$

Figures 3, 4 and 5 were redrawn so that their notation matches these equations (Figure 3), and each caption now describes the figure step by step. Section 4.2 also states how the recommender is combined with the CEFR filter: "Concretely, the candidate items are first restricted to the texts whose CEFR level, estimated with the model of Section 2, matches the level of the learner, possibly after simplification with the model of Section 3; the recommender described below then ranks these candidates according to the interests of the learner. In the offline evaluation of Section 6.3, this level filter is not applied, because the benchmark datasets do not carry CEFR labels: we evaluate the ranking component alone."

> **Reviewer 2 (R2.8):** *"In p. 18, it is claimed that simplified texts "reduce linguistic complexity through careful word choice, syntactic restructuring, and discourse simplification". Is this implicitly achieved through LLM (in this case, the statement is exaggerated), or explicitly guided (how?)?"*

**Response:** It is achieved implicitly via sequence-to-sequence fine-tuning on parallel CEFR pairs rather than explicit syntactic rules. As quoted in Response **AE.4**, we toned down the corresponding

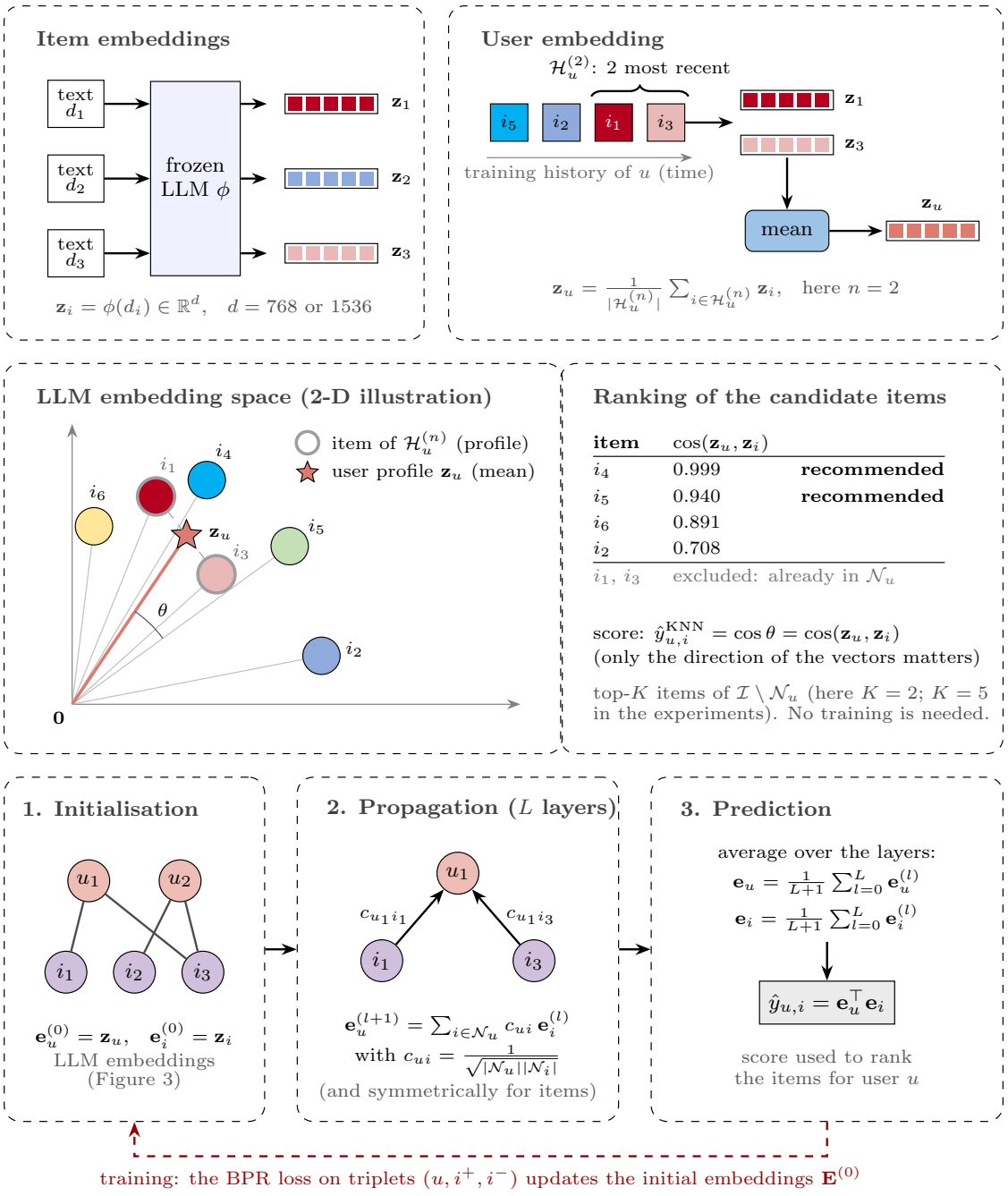



Figure 3: Revised Figures 3, 4 and 5 of the manuscript. *Top* (Figure 3): each item text is encoded by a frozen LLM, and the embedding of a user is the mean of the embeddings of their $n$ most recent training items. *Middle* (Figure 4): content-based KNN; items are scored by their cosine similarity with the user profile $\mathbf{z}_u$. *Bottom* (Figure 5): LightGCN initialised with the LLM embeddings, propagation over $L$ layers, and dot-product score on the averaged embeddings.

sentence in Section 5 (p. 18) accordingly.

> **Reviewer 2 (R2.9):** *"Neither individual components nor the system as a whole are qualitatively evaluated by humans... this is a serious drawback of the study that has to be amended."*

**Response:** As detailed in Response **AE.4** and Table 4, we added Section 6.2.3 reporting a double-blind *Split-and-Overlap* human study on $N = 150$ stratified French sentence pairs (30 per CEFR level A2 to C2, with $n_{\text{shared}} = 50$ overlap pairs annotated by two independent native French evaluators to quantify Cohen's $\kappa$ and Krippendorff's $\alpha$), evaluated with exact two-sided binomial tests on blind preference and paired exact McNemar tests on perceived simplification and meaning preservation rates.

> **Reviewer 2 (R2.10):** *"It is not clear if and how the training loss function incorporates the amplitude of the differences between scores. This is only hinted at on p. 23 but must be much better explained."*

**Response:** We added a dedicated paragraph in Section 6.1.3 (p. 23) clarifying that during fine-tuning, both CamemBERT and the autoregressive LLMs optimize standard categorical cross-entropy loss $\mathcal{L}_{\text{CE}}(x, y) = -\sum_{k=1}^{K} \mathbb{I}(y = k) \log P_\theta(k \mid x)$, which does not penalize ordinal distance $|y - \hat{y}|$ during gradient updates. The concentration of errors within $\pm 1$ adjacent CEFR level in Figure 8 stems from pre-trained semantic representations where adjacent proficiency levels share closer lexical and syntactic distributions. Ordinal distance penalization is enforced at evaluation time via MAE ($|y - \hat{y}|$), QWK ($(y - \hat{y})^2$), and $PM_{\text{norm}}$.

> **Reviewer 2 (R2.11 & R2.12):** *"For misclassifications presented in Fig. 8 and others, compute overall scores such as QWK. None of the evaluations in the tables of Section 6 provides variances and statistical significance of the difference. Please, amend this shortcoming. It seems that some differences are insignificant and therefore the conclusions based on them are inadequate."*

**Response:** As shown directly in the caption and text of Figure 8 and in Table 1 (Response **AE.1**), we computed overall QWK scores for the representative confusion matrices, alongside $B = 1{,}000$ bootstrap standard errors ($\mu \pm \sigma$) and two-sided paired Wilcoxon tests on instance-level ordinal prediction errors with Holm–Bonferroni correction, confirming that misclassifications fall predominantly on adjacent CEFR levels. We also revised Section 6.1.2 to explicitly note non-significant prompt differences on `LjL`. For recommendation (Section 6.3), we now report the mean and standard deviation over 5 random seeds and paired Wilcoxon tests with Holm correction (Tables 2 and 3, Response **AE.1**): the LLM initialisation of LightGCN is not significantly better than the Xavier initialisation on ml-100k, nor with BERT on Zeeguu, and the corresponding sentences of Section 6.3 were corrected. The corrected sentences are listed below (submitted version, then revised version; the rest of the discussion is unchanged):

- Abstract: "showing an 18% increase in NDCG scores" → "showing up to a 17% increase in NDCG scores" (mean over the seeds on Goodreads, +16.7%).
- Section 6.3: "we observe a consistent enhancement in LightGCN's performance when leveraging pre-trained embeddings over Xavier initialization. This improvement is uniform across all four datasets and both embedding methodologies (BERT and ADA)." → "LightGCN initialised with ADA embeddings outperforms the Xavier initialization on Zeeguu, Goodreads and Tomplay (Table 11). With BERT embeddings, the improvement is only significant on Goodreads and Tomplay."
- Section 6.3: "leading to notable gains, particularly in the Zeeguu dataset." → "leading to notable gains, particularly in the Goodreads and Zeeguu datasets." followed by the new sentence "The truncation of the BERT inputs to 128 tokens (Table 7) may explain part of the gap between ADA and BERT, since BERT encodes only 14.5% of the text on Zeeguu, but not all of it: the gap also reflects the difference between the two encoders themselves."
- Section 6.3, Tomplay: "Nonetheless, for both BERT and ADA, pre-trained embeddings outperform the Xavier initialization, reinforcing the superiority of pre-trained embeddings in enhancing recommendation quality across varying dataset characteristics." → "On this dataset, both BERT and ADA embeddings outperform the Xavier initialization."
- Section 6.3, KNN: "while KNN, a purely content-based approach, achieves comparable results to collaborative filtering methods" → "while KNN, a purely content-based approach, remains well below collaborative filtering methods such as ALS".

**Reviewer 2 (R2.13 & R2.14):** *"In iterative simplification (Section 6.2.4), it is not clear what the results of item 3 show, and how they are further processed. Can you improve Fig. 9 by showing where, on average, CEFR levels appear?"*

**Response:** In Section 6.2.2 (formerly Section 6.2.4 in the initial submission), we clarified that at each iteration $t \in \{1, \ldots, 8\}$ across the 100 C2 sentences $x_m^{(0)}$, Step 3 computes the anchor cosine similarity $S_m^{(0\to t)} = \frac{\mathbf{e}(x_m^{(0)}) \cdot \mathbf{e}(x_m^{(t)})}{\|\mathbf{e}(x_m^{(0)})\|_2 \|\mathbf{e}(x_m^{(t)})\|_2}$ against the initial unsimplified sentence $x_m^{(0)}$ to measure cumulative semantic drift. Its mean $\bar{S}^{(t)}$ (red curve in Figure 11 (formerly Figure 9)) remains consistently high across all iterations, staying well above the CamemBERT anisotropy floor $S_0 = 0.81$ and serving as a stopping criterion in our interface whenever $S_m^{(0\to t)} < S_0$. We also added the average CEFR level thresholds in Figure 11 (formerly Figure 9): starting at C2, the mean trajectory crosses C1 at $t = 1$, crosses B2 at $t = 2$, and reaches an asymptotic plateau near the B1 boundary across later iterations.

**Reviewer 2 (R2.15 & R2.16):** *"The metric NDCG@5 shall be explained in the text prior to its first use. On p. 30 (or before), better explain Xavier initialization and the proposed improvement."*

**Response:** In Section 6.3.3, before its first mention, we added the formal definitions of $\mathrm{DCG}_u@K$, $\mathrm{IDCG}_u@K$, and $\mathrm{NDCG}@K$ (for $K = 5$), followed by the Glorot (Xavier) uniform initialization distribution:

$$\begin{gathered} \mathrm{DCG}_u@K = \sum_{r=1}^{K} \frac{2^{\mathrm{rel}_{u,\pi_u(r)}} - 1}{\log_2(r+1)}, \qquad \mathrm{NDCG}@K = \frac{1}{|\mathcal{U}_{\text{test}}|} \sum_{u \in \mathcal{U}_{\text{test}}} \frac{\mathrm{DCG}_u@K}{\mathrm{IDCG}_u@K}, \\ e_{j,m}^{(0)} \sim \mathcal{U}\left( -\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, +\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}} \right). \end{gathered} \tag{8}$$

where $n_{\text{in}} + n_{\text{out}}$ is $|\mathcal{U}| + d$ for the user embeddings and $|\mathcal{I}| + d$ for the item embeddings. We explain that Xavier initialization assigns random, semantics-free layer-0 vectors, which suffers under extreme interaction sparsity (`Zeeguu` 99.92%, `Goodreads` 99.99%). Replacing random draws with pre-trained LLM embeddings $\mathbf{e}_i^{(0)} = \phi_{\text{LLM}}(d_i)$ clusters semantically related texts at layer 0 prior to BPR optimization, yielding significant NDCG@5 improvements on `Zeeguu` and `Goodreads` with ADA (means over 5 random seeds, significant after Holm correction, Table 3).

**Reviewer 2 (R2.17):** *"[At the end] of Section 6.3.4 (or later) provide recommendations on what components to use."*

**Response:** Across Sections 6.1, 6.2, and the Conclusion (Section 7), we synthesize our component recommendations ((1) for **Difficulty Estimation**, prompted **GPT-4o-mini** achieves the best accuracy-cost trade-off while fine-tuned **CamemBERT** minimizes severe ordinal errors; (2) for **Text Simplification**, LoRA fine-tuned **Mistral-7B** achieves high simplification accuracy and semantic preservation without over-condensing content (Table 4 and Table 5)); and at the end of Section 6.3.4 we added concrete deployment guidelines for (3) **Recommendation**, revised with the 5-seed results: (i) on text-rich, sparse catalogues (`Zeeguu`, `Goodreads`), initialise LightGCN with the 1,536-dimensional ADA embeddings ($L = 9$ layers on `Zeeguu`, $L = 2$ on `Goodreads`, user profiles of $n \in \{1, 2\}$ items), which yields statistically significant NDCG@5 improvements over the Xavier initialisation (Table 11 and Table 3); (ii) on structured metadata catalogues (`Tomplay`, `ml-100k`), use the 768-dimensional multilingual BERT embeddings, which outperform ADA on `Tomplay`, whose texts are short enough to be encoded entirely, and are not significantly different from ADA on `ml-100k`, at no inference cost; (iii) when graph training is not possible (latency or cold-start constraints), use the content-based KNN over ADA embeddings, with a user profile made of the $n = 1$ to 4 most recent items, as a lightweight fallback.

**Reviewer 2 (R2.18):** *"The claims of a language-agnostic approach on p. 33 (and elsewhere) are somewhat exaggerated, as many of the used components are French-specific. The approach's findings are useful for other languages, but the formulation shall be adapted."*

**Response:** We revised the Abstract, Section 1, and Section 7 to state explicitly that our empirical

validation, corpora, and CamemBERT encoder are French-specific, while the methodological architecture ($PM_{\text{norm}}$, $A_K/S$, and LLM-initialized LightGCN) can be transferred to other languages by substituting target-language encoders and CEFR-labeled training pairs.

> **Reviewer 2 (R2.19–R2.22):** *"Some of the links provided in the paper do not work. The prompt on p. 20 is repeated from p. 8. Remove it and provide a reference. On p. 29, there are missing references (??). References must be improved (replace arXiv with published variants, remove redundant conference fields, capitalize acronyms)."*

**Response:** We addressed all four minor points: (1) verified all external URLs across the manuscript and updated the broken external video links in Section 5 and Section 3 to point to the project repository and direct video demonstration (https://github.com/hjamet/LLM-Difficulty-Based-Recommendation/blob/main/demos/lingorank_simplification_demo.mp4); (2) restored Section 2.2.4 and replaced the duplicate quote block in Section 6 (p. 20) with a cross-reference to Section 2.2.4; (3) resolved the ?? on p. 29 so that it points cleanly to Table 9 and Table 12 (formerly Table 9 in the initial submission) in the Appendix; and (4) cleaned the bibliography: every cited entry now gives its source (its DOI, or the official URL when there is none), arXiv preprints are replaced by their published versions when they exist (for example, BPR at UAI 2009, PEGASUS at ICML 2020, Imperial at RANLP 2021, Xia et al. at BEA 2016, Xu et al. in ACM TKDD), the ten preprints without a published version are kept as arXiv references with their arXiv DOI, conference entries no longer contain editors, locations, publishers or dates, acronyms are protected (such as LLM, BERT, GPT, CEFR, LightGCN), and incorrect entries were corrected (missing or wrong authors, wrong DOIs, a duplicate entry, and Tuncay (2020) verified and updated to its official Master's thesis record on McGill eScholarship).

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